

Data analysis in the frequency domain: notes

Marco Vignati

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Chapter 1

Fourier series and transform

1.1 Fourier series

In 1807 Joseph Fourier proposed to decompose complicated functions into sums of sine and cosine waves, in order to find solutions of equations for the propagation of heat. The idea is based on the fact that sine and cosine waves and their harmonics represent a complete set of eigenfunctions and therefore an arbitrary¹ function can be represented as a linear combination of infinite eigenfunctions with appropriate coefficients. The coefficients are the projection of the function on a sine or cosine harmonic.

In the Fourier's analysis a periodic function $f(t)$ with period T is decomposed in a discrete sum of harmonics with fundamental angular frequency $\omega_0 = \frac{2\pi}{T}$ as:

$$f(t) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos(k\omega_0 t) + b_k \sin(k\omega_0 t) \quad (1.1)$$

with

$$a_k = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \cos(k\omega_0 t) dt \quad (1.2)$$

$$b_k = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \sin(k\omega_0 t) dt \quad (1.3)$$

We note that:

- $a_0/2$ is the mean value of $f(t)$;

¹Arbitrary in Physics and Engineering problems, not as arbitrary as Mathematicians can think

- $b_0 = 0$ by definition;
- ω_0 is the lowest harmonics, components with smaller frequency are packed in a_0 ;
- If $f(t)$ originally is not periodic, the Fourier sum is the repetition of $f(t)$ in time with period T . This implies that non-periodic functions becomes periodic when re-built with the Fourier series.

The Fourier series in Eq. [1.1](#) can be represented in the complex plane by substituting the cos and sin functions with the Euler's formulas:

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}, \quad \sin x = \frac{e^{ix} - e^{-ix}}{2i} \quad (1.4)$$

$$f(t) = c_0 + \sum_{k=1}^{\infty} c_k e^{ik\omega_0 t} + c_k^* e^{-ik\omega_0 t} = \sum_{k=-\infty}^{\infty} c_k e^{ik\omega_0 t} \quad (1.5)$$

where

$$c_k = \frac{1}{2} (a_k - ib_k) = \frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-ik\omega_0 t} dt \quad (1.6)$$

1.1.1 Example: square wave

In the following example we compute the series coefficients of a square wave of period T with amplitude 0 in the interval $[-T/2, 0]$ and amplitude V in the interval $[0, T/2]$:

$$c_k = \frac{V}{T} \int_0^{T/2} e^{-ik\omega_0 t} dt \quad (1.7)$$

$$c_k = \frac{V}{T} \frac{1}{ik\omega_0} (1 - e^{-ik\pi}) \quad (1.8)$$

So

$$c_0 = \lim_{k \rightarrow 0} c_k = \frac{V}{2} \quad (1.9)$$

$$c_k = 0, \quad \text{even values of } k \quad (1.10)$$

$$c_k = \frac{V}{\pi} \frac{1}{ik}, \quad \text{odd values of } k \quad (1.11)$$

Considering that $c_{-k} = -c_k$, from Eq. [1.5](#) we can write the series as

$$f(t) = \frac{V}{2} + \frac{2V}{\pi} \left[\frac{\sin(\omega_0 t)}{1} + \frac{\sin(3\omega_0 t)}{3} + \frac{\sin(5\omega_0 t)}{5} + \dots \right] \quad (1.12)$$

1.2 Fourier transform

Non-periodic functions, like for example a pulse, imply that $T \rightarrow \infty$. In this limit the spacing between consecutive harmonics,

$$\Delta\omega = \omega_0 = \frac{2\pi}{T} \quad (1.13)$$

tends to zero. It is therefore more appropriate to think about density of waves rather than to waves with specific frequency. Indeed from equation [1.6](#) we would obtain $c_k \rightarrow 0$, i.e. the contribution of a single wave with specific frequency $\omega = k\omega_0$ is null. This density can be obtained as:

$$g(\omega) = \mathcal{F}[f(t)] = \lim_{T \rightarrow \infty} T c_k = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt \quad (1.14)$$

which is the so called “Fourier transform”. The anti-transform to obtain back $f(t)$ can be obtained from Eq. [1.5](#) as

$$f(t) = \mathcal{F}^{-1}[g(\omega)] = \lim_{T \rightarrow \infty} \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} T c_k e^{ik\omega_0 t} \Delta\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} g(\omega) e^{i\omega t} d\omega \quad (1.15)$$

The following properties of the Fourier transform can be derived from Eq. [1.14](#):

- Conjugation (only if $f(t)$ is real): $g(-\omega) = g^*(\omega)$
- Time shift: $\mathcal{F}[f(t - a)] = g(\omega) e^{-i\omega a}$
- Convolution: $\mathcal{F}[f(t) * g(t)] = f(\omega) \cdot g(\omega)$
- Parseval’s identity: $\int_{-\infty}^{\infty} |f(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |g(\omega)|^2 d\omega$

Useful pairs are

- $\mathcal{F}[\delta(t)] = 1$
- $\mathcal{F}[1] = 2\pi\delta(\omega)$
- $\mathcal{F}[\theta(t)e^{-t/\tau}] = \frac{\tau}{1+i\tau\omega}$

1.2.1 Example: square pulse

In the following example we compute the Fourier Transform of a square pulse with amplitude V in the interval $[0, T/2]$ and zero elsewhere:

$$g(\omega) = \int_0^{T/2} V e^{-i\omega t} dt \quad (1.16)$$

and the result is:

$$g(\omega) = \frac{V}{i\omega} \left(1 - e^{-i\omega T/2}\right) = \frac{2V}{\omega} e^{-i\omega T/4} \sin \frac{\omega T}{4} \quad (1.17)$$

The magnitude is:

$$|g(\omega)| = \frac{2V}{\omega} \sin \frac{\omega T}{4} \quad (1.18)$$

to be compared with the coefficients of the series:

- $|g(k\omega_0)| = \frac{VT}{\pi k} \left| \sin \frac{k\pi}{2} \right| = c_k \cdot T$ and the mean (c_0) becomes the integral.
- $|g(\omega)| = |g(-\omega)|$, in the series we have ($c_k = c_{-k}$)

The phase is:

$$\arg[g(\omega)] = -\frac{\omega T}{4} \quad (1.19)$$

with:

- $\arg[g(k\omega_0)] = -k\frac{\pi}{2}$

1.2.2 Example: Heaviside $\theta(t)$

The Heaviside function is of particular importance in the description of physical processes, as it allows us to describe the start time of an event in the math of signal processing.

$$\theta(t - t_0) = \begin{cases} 1 & \text{if } t \geq t_0 \\ 0 & \text{if } t < t_0 \end{cases} \quad (1.20)$$

The transform of $\theta(t)$ reads:

$$g(\omega) = \int_0^{\infty} e^{-i\omega t} dt = \frac{1}{i\omega} (1 - e^{-i\omega\infty}) \quad (1.21)$$

Who knows what is the value of an oscillatory function at $t = \infty$? The apparently does not converge, one has to force the convergence by dumping the oscillations:

$$g(\omega) = \lim_{\sigma \rightarrow 0} \int_0^{\infty} e^{-\sigma t - i\omega t} dt = \lim_{\sigma \rightarrow 0} \frac{1}{i\omega + \sigma} \quad (1.22)$$

where $\sigma > 0$ is a real number. By computing the limit we obtain:

$$g(\omega) = \lim_{\sigma \rightarrow 0} \frac{1}{i\omega + \sigma} = \frac{1}{i\omega} + \lim_{\sigma \rightarrow 0} \frac{\sigma}{\omega^2 + \sigma^2} = \frac{1}{i\omega} + \pi\delta(\omega) \quad (1.23)$$

1.3 Laplace Transform

The Laplace Transform replaces $i\omega$ with $s = i\omega + \sigma$ to ensure convergence of the Fourier transform, and restricts the integration to positive times, that is useful to describe transient signals as in physical systems. Analog filters are usually designed and expressed in the Laplace domain. The Laplace transform is defined as:

$$\mathcal{L}[f(t)] = F(s) = \int_{0^-}^{\infty} f(t)e^{-st} dt, \quad s = \sigma + i\omega \quad (1.24)$$

where the 0^- is intended to capture the entire $\delta(t)$. Useful pairs are:

- $\mathcal{L}[\delta(t)] = 1$
- $\mathcal{L}[\theta(t)] = \frac{1}{s}$
- $\mathcal{L}[e^{-t/\tau}] = \frac{\tau}{1+s\tau}$
- $\mathcal{L}[\int_0^t f(t') dt'] = \frac{1}{s} F(s)$
- $\mathcal{L}[\frac{df(t)}{dt}] = sF(s) - f(0^-)$

After performing calculations in the Laplace domain, quantities in the (physical) frequency domain can be obtained by taking the limit:

$$\lim_{\sigma \rightarrow 0} s = i\omega \quad (1.25)$$

One of the advantages of this transform is that differential equations becomes algebraic equations, and that the initial conditions are directly plugged in (not shown in this note, but very useful to solve systems of differential equations describing electric or thermal circuits).

Chapter 2

Discrete time

Data acquisition systems are based on analog to digital converters (ADC). In an ADC, an analog signal is sampled at regular time intervals Δt and each sample is converted into a binary (digital) number. The ADC is then complemented by a memory where the samples are written for subsequent reading. The fundamental parameters characterizing an ADC are the sampling frequency ($f_s = 1/\Delta t$), the number of bits M and the dynamic range of the accepted analog signal ΔV . The higher are f_s and M , the higher is the cost.

As in an oscilloscope, the analog or the digital signal is triggered, in order to read from memory a bunch of samples, i.e a waveform. A waveform is therefore a digitized snapshot the analog signal of N samples, where N is chosen by the user. In this chapter we will introduce the math to analyze discrete and finite time sequences.

2.1 Nyquist theorem and aliasing

The theorem states that to represent a wave with frequency f , a sampling frequency $f_s \geq 2f$ is needed. Conversely, we can state that given a sampling frequency f_s , the maximum frequency that can be represented is the Nyquist frequency:

$$f_N = \frac{f_s}{2} \quad (2.1)$$

Waves with frequency $\geq f_N$ will be represented as lower frequency waves, an effect dubbed “aliasing”. The reconstructed frequency f_r from the original frequency f is

$$f_r = |2mf_N - f| \quad \text{with } m \text{ such that } f_r < f_N \quad (2.2)$$

2.2 Discrete Fourier Transform

The Fourier series in Sec. 1.1 represents the decomposition of a continuous periodic function into a sum of waves with discrete angular frequencies. In case of a discrete signal f_n sampled at fixed time intervals Δt for $N = \frac{T}{\Delta t}$ times, the coefficients in Eq. 1.6 can be computed as

$$X_k = \frac{1}{N} \sum_{n=-(N/2-1)}^{N/2} f_n e^{-i \frac{2\pi k}{N} n} \quad (2.3)$$

From the above equation it can be seen that

$$X_{N+k} = X_k \quad (2.4)$$

i.e. the transform is periodic in N and therefore only N complex numbers can be squeezed out from f_n (which is indeed also periodic in N). Note that the terms $n = 0$ (zero frequency, mean value of f_n) and $n = N/2$ (Nyquist frequency) are real, i.e. it is not possible to associate a phase to them (well, the real part contains the sign, which in case of a wave correspond to phase π). Moreover if f_n is real the following property holds:

$$X_{-k} = X_k^* \quad (2.5)$$

The frequency resolution is

$$\Delta\omega = \frac{2\pi}{T} \quad (2.6)$$

which also represents the lowest frequency that can be measured.

The anti-transform, i.e. the “Fourier series of a discrete signal” is (see Eq. 1.5):

$$f_n = \sum_{k=-(N/2-1)}^{N/2} X_k e^{i \frac{2\pi k}{N} n} \quad (2.7)$$

2.3 Discrete Fourier Transform in computers

DFT algorithms implemented in computer libraries use a different conventions for efficiency reasons. DFT uses $O(N^2)$ operations, while the Fast Fourier Transform (FFT) use $O(N \log(N))$ operations. The algorithms use the periodicity property

$$X_{-k} = X_{N-k} \quad (2.8)$$

to map negative indices into positive indices. In this representation negative frequencies are listed in the second half of the sequence and in reverse order, i.e. the lower the latter. Additionally it removes the $1/N$ operation in Eq. [2.3](#). The FFT pairs are therefore:

$$X_k = \sum_{n=0}^{N-1} f_n e^{-i \frac{2\pi k}{N} n} \quad (2.9)$$

$$f_n = \frac{1}{N} \sum_{k=0}^{N-1} X_k e^{i \frac{2\pi k}{N} n} \quad (2.10)$$

Others implementations in place of the $(1, 1/N)$ normalizations use $(1/\sqrt{N}, 1/\sqrt{N})$ so be warned! Finally, in this representation if f_n is real the following property holds:

$$X_{N-k} = X_k^* \quad (2.11)$$

2.4 Example: DFT of a cosine

Let's consider a cosine wave with frequency index $k = k'$

$$f_n = \cos\left(\frac{2\pi k'}{N} n\right) \quad (2.12)$$

The DFT reads:

$$X_k = \sum_{n=0}^{N-1} \frac{e^{i \frac{2\pi k'}{N} n} + e^{-i \frac{2\pi k'}{N} n}}{2} e^{-i \frac{2\pi k}{N} n} \quad (2.13)$$

$$X_k = \sum_{n=0}^{N-1} \frac{e^{i \frac{2\pi (k' - k)}{N} n} + e^{-i \frac{2\pi (k' + k)}{N} n}}{2} \quad (2.14)$$

The sum of each term over the entire circle is zero. Only sums with $k' - k = 0$ or $k' + k = 0$ will contribute:

$$X_k = \begin{cases} \frac{N}{2} & \text{if } k = k' \\ \frac{N}{2} & \text{if } k = -k' = N - k' \\ 0 & \text{otherwise} \end{cases} \quad (2.15)$$

Note that in this special case X_k is real.

2.5 Example: DFT of a sine

Let's consider a sine wave with frequency index $k = k'$

$$f_n = \sin\left(\frac{2\pi k'}{N}n\right) \quad (2.16)$$

The DFT reads:

$$X_k = \sum_{n=0}^{N-1} \frac{e^{i\frac{2\pi k'}{N}n} - e^{-i\frac{2\pi k'}{N}n}}{2i} e^{-i\frac{2\pi k}{N}n} \quad (2.17)$$

So

$$X_k = \begin{cases} \frac{N}{2i} & \text{if } k = k' \\ -\frac{N}{2i} & \text{if } k = -k' = N - k' \\ 0 & \text{otherwise} \end{cases} \quad (2.18)$$

So the magnitude of the coefficients is the same as in the cos transform, but they are purely imaginary with phase $\pi/2$ and $-\pi/2$

2.6 Windowing: see slides

Chapter 3

Noise

3.1 Theory

The noise is a stochastic process which is often assumed to be stationary, i.e. the statistical properties do not vary with time, and ergodic, i.e. the statistical properties can be extracted from the observation of a single realization. In these conditions the first-order probability function to describe the process does not depend on time, and in most cases can be assumed Gaussian. Since we are interested in studying fluctuations only, the description of the noise usually is that of process with zero mean:

$$P[f(t)] = \frac{1}{2\pi\sqrt{\sigma}} e^{-f^2(t)/\sigma^2} \quad (3.1)$$

So at the first order the parameter describing the noise is:

$$\sigma^2 = E[f^2(t)] = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} f^2(t) dt \quad (3.2)$$

which can be interpreted as the total power of the noise process.

At the second-order, the process can be described by its time autocorrelation:

$$R(\tau) = E[f(t+\tau)f(t)] = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} f(t+\tau)f(t) dt \quad (3.3)$$

which describes the degree of prediction of $f(t+\tau)$ when $f(t)$ is known. For $\tau = 0$, $R(0) = \sigma^2$.

The power spectral density is defined as the Fourier transform of $R(\tau)$ according to the Wiener-Khinchin theorem:

$$S^D(\omega) = \int_{-\infty}^{\infty} R(\tau) e^{-i\omega\tau} d\tau \quad (3.4)$$

$$R(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S^D(\omega) e^{i\omega\tau} d\omega \quad (3.5)$$

From the last equation we can derive:

$$\sigma^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} S^D(\omega) d\omega \quad (3.6)$$

where we can see that the power spectral density is the contribution to the variance of $f(t)$ given by an unitary interval of waves at frequency ω . From the above equation we can estimate it using the Parseval's identity:

$$\sigma^2 = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} f^2(t) dt = \lim_{T \rightarrow \infty} \frac{1}{T} \frac{1}{2\pi} \int_{-\infty}^{\infty} |g(\omega)|^2 d\omega \quad (3.7)$$

so that

$$S^D(\omega) = \lim_{T \rightarrow \infty} \frac{|g(\omega)|^2}{T} \quad (3.8)$$

It is now interesting to look at the simplest form of noise, the so called “white” noise:

$$S^D(\omega) = k \quad (3.9)$$

which has an equal contribution at all frequencies (and hence the name). From Eq. 3.5 the autocorrelation is $R(\tau) = k\delta(\tau)$ and this implies that $\sigma^2 = \infty$! We therefore must define a bandwidth $\Delta\omega$ to limit the integration of Eq. 3.5, then $\sigma^2 = \frac{2k\Delta\omega}{2\pi}$.

In most cases people refer to the single-side-band spectrum, considering only positive frequencies. Since negative frequencies have identical contributions of positive ones it is sufficient to restrict the integral to positive frequencies and multiply by a factor of 2 (remember that this applies only to fluctuations, not to the mean value):

$$S(\omega) = 2\theta(\omega)S^D(\omega) \quad (3.10)$$

3.2 Shot

The average current traversing a barrier in a conductor is:

$$I = q\Gamma \quad (3.11)$$

where Γ is the rate of passage of particles with charge q through the barrier. We can describe the current in time as a Poisson process with random arrival time t_j of the j -th particle:

$$I(t) = \lim_{T \rightarrow \infty} \sum_{j=-T\Gamma/2}^{T\Gamma/2} q\delta(t - t_j) \quad (3.12)$$

where remember that the $\delta(t)$ has dimension of t^{-1} since the integral is 1 by definition. Since

$$\mathcal{F}[\delta(t - t_j)] = e^{-i\omega t_j} \quad (3.13)$$

from Eq. [3.8](#) and Eq. [3.10](#) we have:

$$S_I(\omega) = \lim_{T \rightarrow \infty} \frac{2q^2}{T} \left[\sum_j e^{-i\omega t_j} \sum_k e^{+i\omega t_k} \right] \quad (3.14)$$

Let's focus on the terms with $\omega \neq 0$. If $j \neq k$ the sum averages to zero. Instead if $j = k$, $\sum_j 1 = \Gamma T$ and thus

$$S_I(\omega) = 2qI \quad (3.15)$$

The terms with $\omega = 0$ return the square of the mean current, and are not noise.

3.3 Johnson-Nyquist

The Johnson-Nyquist (or simply Johnson) noise is the noise generated by the thermal energy of a resistor at temperature T . To compute this noise we will follow the demonstration done by Nyquist in 1927, which exploits the thermal equilibrium of two resistors that are at same temperature and connected by a wire ([Fig. 3.3](#)). Because of equilibrium, a continuous power flows from one resistor to the other and viceversa via elctro-magnetic waves. In the following we will compute the number of theses waves, their energy and the power transferred. Because of symmetry there is not a net energy

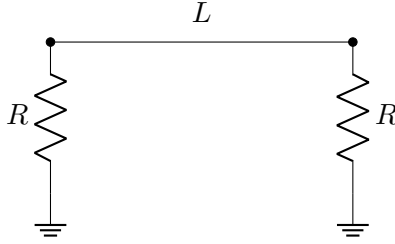


Figure 3.1: Circuit for the calculation of the Johnson-Nyquist noise: two identical resistors at same temperature T are connected by an ideal wire.

transfer from a resistor to the other, this means that the combination of the two opposite waves (from left to right and from right to left) should give rise to a standing wave, e.g. a wave with fixed nodes, like in a guitar.

The lowest frequency allowed is $\nu = c/2L$, i.e. a wave with wavelength $2L$. In general all harmonics m of this wave are allowed:

$$\nu = \frac{c}{2L}m \quad (3.16)$$

Considering that there is no upper limit to m , we can compute the density of modes per unit frequency:

$$\frac{dm}{d\nu} = \frac{2L}{c} \quad (3.17)$$

Each wave m is a degree of freedom of the system, and carries an average energy $k_B T$. The energy density per unit frequency is therefore

$$\frac{dE}{d\nu} = \frac{2L}{c} k_B T \quad (3.18)$$

The power of each resistor can be obtained by noticing that the travelling time is L/c and that each resistor contributes $1/2$:

$$\frac{dP}{d\nu} = k_B T \quad (3.19)$$

which represents the power spectrum in $[\text{W/Hz}]$ units contributed by a resistor and which is white, since each wave has an arbitrary phase. The voltage noise spectral density can be computed as

$$P = I^2 R = \frac{V^2}{4R^2} R \quad \longrightarrow \quad S_V(\nu) = 4k_B T R \quad (3.20)$$

3.4 Generation-Recombination

Generation-recombination noise is typical of semi-conductors, where in temperature equilibrium an electron-hole pair can be created and then recombine. The same can happen in superconductors, where Cooper pairs can be broken and then recombine. This phenomenon can be described as a superposition of signals of amplitude A , where A is the measurable variable (e.g. voltage or current). The signals last on average τ_{rec} and occur in time as a Poissonian process with rate Γ .

Similarly to the calculation of the shot noise we can write

$$A(t) = \lim_{T \rightarrow \infty} \sum_{j=-T\Gamma/2}^{T\Gamma/2} A e^{-\frac{t-t_j}{\tau_{\text{rec}}}} \quad (3.21)$$

Considering that $\mathcal{F}[e^{-t/\tau}] = \frac{\tau}{1+j\omega\tau}$:

$$S_{GR}(\omega) = \lim_{T \rightarrow \infty} \frac{2A^2}{T} \frac{\tau_{\text{rec}}^2}{1 + \omega^2 \tau_{\text{rec}}^2} \left[\sum_j e^{-i\omega t_j} \sum_k e^{+i\omega t_k} \right] \quad (3.22)$$

we have that

$$S_{GR}(\omega) = \frac{2A^2 \Gamma \tau_{\text{rec}}^2}{1 + \omega^2 \tau_{\text{rec}}^2} \quad (3.23)$$

where the shape is the same as white noise filtered by a single-pole RC filter.

3.5 Flicker

Flicker noise has a spectrum

$$S_{\text{flicker}}(\omega) \propto \frac{1}{\omega^\alpha} \quad \text{with } \alpha \simeq 1 \quad (3.24)$$

Unlike the previous sources of noise, Flicker noise has not a clear origin or explanation. However it is almost always present in electronic devices, detectors, but also in biological systems. A possible explanation is that it originates from a superposition of generation-recombination events with τ_{rec} randomly distributed. Interestingly, a spectrum with $\alpha = 1$ is obtained from Eq. [3.21](#) with a uniform distribution applied to the cutoff angular frequency $1/\tau_{\text{rec}}$.

3.6 DFT estimation of the power spectral density

The power of each complex wave can be computed as

$$P_k = \frac{X_k^2}{N^2} \quad (3.25)$$

Then remember that each frequency (not angular frequency) has width $\Delta f = 1/T$, so the power spectral density is:

$$S_k = T \frac{X_k^2}{N^2} = \frac{X_k^2}{N f_s} \quad (3.26)$$

To obtain a good estimate of S_k , M noise waveforms are acquired and then averaged:

$$\hat{S}_k = \frac{\sum_{l=1}^M S_k^l}{M} \quad (3.27)$$

3.7 Quantization noise

Quantization occurs when an analog signal (or noise) is acquired with an ADC. The analog value of the signal is rounded (or truncated) to the nearest digital number. The error is that of the uniform distribution

$$\sigma_Q^2 = \frac{1}{3} \quad (3.28)$$

and the error of each sample is independent from the others, meaning that the noise is white:

$$S_Q(\omega) = 2\sigma_Q^2 \frac{2\pi}{\Delta\omega} \quad (3.29)$$

where $\Delta\omega$ is the Nyquist angular frequency of the ADC. Therefore, for same number of bits, the higher the bandwidth the lower the noise density.